

Reading

Let us start reading the data we downloaded. First of all, we have to create a new notebook called *Open Government Data Analysis* and open it. Then, after ensuring that the `educ_figdp_1_Data.csv` file is stored in the same directory as our notebook directory, we will write the following code to read and show the content:

```
edu = pd.read_csv('files/od02/educ_figdp_1_Data.csv',
                  na_values = '.',
                  usecols = ["TIME", "GEO", "Value"])
edu
```

	TIME	GEO	Value
0	2000	European Union ...	NaN
1	2001	European Union ...	NaN
2	2002	European Union ...	5.00
3	2003	European Union ...	5.03
...	...	...	...
382	2010	Finland	6.85
383	2011	Finland	6.76

384 rows x 4 columns

The way to read CSV (or any other separated value, providing the separator character) files in Pandas is by calling the `read_csv` method. Besides the name of the file, we add the `na_values` key argument to this method along with the character that represents "non available data" in the file. Normally, CSV files have a header with the names of the columns. If this is the case, we can use the `usecols` parameter to select which columns in the file will be used.

In this case, the DataFrame resulting from reading our data is stored in `edu`. The output of the execution shows that the `edu` DataFrame size is 384 rows x 4 columns. Since the DataFrame is too large to be fully displayed, three dots appear in the middle of each row.

Beside this, Pandas also has functions for reading files with formats such as Excel, HDFS, tabulated files, or even the content from the clipboard (`read_excel()`, `read_hdf()`, `read_table()`, `read_clipboard()`). Whichever function we use, the result of reading a file is stored as a DataFrame structure.

To see how the data looks, we can use the `head()` method, which shows just the first five rows. If we use a number as an argument to this method, this will be the number of rows that will be listed: